Amira Botros Enaam Shehata Henin

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PROFILE

Motivated undergraduate engineering student with strong programming skills and fluency in English. Experienced in international exchange through a fully funded scholarship program in the United States. Proven ability to work collaboratively in team environments and communicate effectively across diverse settings. Passionate about continuous learning and personal growth, with a strong drive to excel academically and professionally in the field of engineering.

EDUCATION

Bachelor of Computer Engineering and Artificial Intelligence 10/2024 – Present
Faculty of Engineering Ain Shams (Credit Hour System)
Dual degree with the University of East London

IGCSE 09/2022 – 06/2024
Al Karma International School
Grade: 99.7%

TRAINING EXPERIENCE

Coach Academy for Competitive Programming
Competitive Programming Trainee
- Completed a 30-hour intensive training program over two weeks, covering competitive programming fundamentals and problem-solving techniques
- Practiced algorithm design, data structure implementation, and timed problem solving under competition conditions

SKILLS

Programming Languages — C++, Java, Python, C#, SQL, C, VHDL, Arduino
Software & Design — UML Design, Software Engineering, Object-Oriented Programming, Logic Design & Computer Organization
Tools & Simulation — Matlab, Simulink
Soft Skills — Problem Solving, Team Collaboration, Fast Learning

LANGUAGES

Arabic
Native Language

English
SAT - (1110)

French
Delf A2

CERTIFICATES

Yes Scholarship
A year-long exchange program provided by the U.S. Department Of State to study for a year in the United
States and live with an American host family

Outstanding Achievements in Gymnastics
Winning 12 gold, 6 silver, and 3 bronze medals in national championships

PROJECTS

FOS — Operating System Kernel
- Extended an educational OS kernel in C, implementing core memory management and hardware-level resource allocation
- Developed dynamic kernel heap placement strategies and custom page file handling algorithms for efficient memory allocation
- Architected a multi-level feedback queue for efficient, priority-based CPU process scheduling

AI Scene Style Classification Model
- Collaborated in developing a deep learning model for scene style classification
- Preprocessed image datasets and utilized pretrained backbone architectures for feature extraction
- Designed a custom classification head and applied ensemble techniques to improve prediction accuracy

- Evaluated model performance using machine learning metrics and validation techniques

Cinema Management System
- Designed and developed a relational cinema database using SQL, including schema creation, normalization, and database relationships
- Implemented SQL queries, stored procedures, and functions to manage bookings, screenings, users, and ticket operations
- Developed a graphical user interface (GUI) using C# and integrated it with the SQL database to support real-time system functionality and data management

TINY Language Compiler
- Architected a complete language compiler from scratch in C# for a custom specification, covering all major compilation phases
- Built a lexical analyzer and recursive descent parser to generate Abstract Syntax Trees supporting loops, conditionals, and functions

Tripzy — Travel Booking Platform
- Collaborated within a 10-person team on a full-stack role-based travel booking platform using Next.js, TypeScript, Tailwind CSS, and Firebase
- Contributed to establishing Git branching workflows that resolved integration bottlenecks across the full stack
- Helped architect real-time Firestore data synchronisation, dynamic pricing logic, and a fully responsive Tailwind CSS UI

MIPS Processor Design using VHDL
- Contributed to the design and implementation of a basic MIPS processor using VHDL
- Developed and tested instruction execution modules for arithmetic, logic, and memory operations
- Participated in simulation and debugging to verify processor functionality and instruction flow

Travel Booking System
- Conducted system analysis and software design, including requirements gathering, UML diagrams, and database design
- Developed and implemented booking, reservation, and user management functionalities

Min and Max Heap using C++
- Implemented core heap operations, including insertion, deletion, heapify, extract-min/max, and priority queue functionality
- Analyzed heap operation time complexities and optimized performance using efficient data structures

VOLUNTEERING EXPERIENCE

Volunteer with AFS Egypt
- Hosted exchange students from the UK, China, and Italy, fostering cross-cultural understanding and building long lasting international connections.

Volunteer with St. Arsanios Education 2024 – present
- Provided free education in English, Arabic, Mathematics, and essential life skills to underprivileged students who lack access to quality learning.
- Made a measurable impact by equipping children with both academic knowledge and practical skills, including personal hygiene and self-development, to improve their long-term opportunities.

Volunteer with El Rae El Saleh 2024 – present
- Helped in the process of preparing thousands of food boxes for needy families

REFERENCES

Available Upon Request